

# Pricing the DeFi Tail: Do Protocols or Depositors Price Operational Risk?

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**Abstract.** Similar to banks, DeFi protocols expose depositors to operational risk (USD 9.45 billion across 1,075 events since 2020). Unlike banks, they are not required to hold capital against it. A protocol may maintain a buffer voluntarily. Absent one, the risk falls on the depositor, who should then demand a risk premium in the supply yield. I quantify the underlying tail on one benchmark, a per-sector Basel loss-distribution approach fitted to a new operational risk event dataset, and test both margins against it. Tails in the four core sectors are no heavier than the Moscadelli banking band [0.85, 1.39]. Bridge, Derivatives, and the residual Other sector exhibit cyber-loss-level tails ( $\xi \approx 1.6$ ), with point estimates past the infinite-mean boundary. The Lending tail implies a  $\text{VaR}_{99.9}$  capital buffer of 18% of TVL and of the ten largest Lending venues, the four holding a buffer cover on average 5% of it. Under market discipline, depositors should demand a higher yield in compensation where a venue does not maintain a buffer. I find that venues without a buffer pay a higher premium than those with (a 125-bps gap in medians): evidence the market discriminates in the right direction. However, the premium falls far short of an adequately priced tail. This unpriced tail falls disproportionately on the retail depositor, who sees only the posted rate but lacks the information and skills to price it. Because these products are not bank-regulated, I recommend disclosure over capital mandates: protocols, and any service providers that front access to it, should publish standardized losses, existing capital buffers and tail coverage.

**Keywords:** operational risk · extreme value theory · loss-distribution approach · capital requirements · decentralized finance · market discipline

## 1 Introduction

Decentralized finance (DeFi) offers yield-bearing *earn products*, among them lending-pool supply positions, decentralized-exchange (DEX) liquidity-provision positions, and yield vaults. Similar to a bank deposit, they expose the depositor to *operational risk*, the losses from failed processes, people, systems, and external events. Unlike a bank, which must hold capital against operational risk sized under Basel from a *loss-distribution approach* (LDA) [9, 10, 19, 22], a DeFi protocol faces no such requirement, so no regulated buffer stands between a loss event and the depositor's claim.

DeFi’s architecture genuinely removes several bank-balance-sheet risks: smart-contract custody replaces the custodian, settlement is atomic, and counterparty credit risk collapses into the overcollateralization embedded in the contract [3, 30]. But the operational risks do not go away, and in some cases concentrate: external attacks on contract code, insider-rugpull events, oracle and economic-design manipulation, and configuration errors. Aramonte et al. [1] call this the “decentralisation illusion.” The consolidated dataset assembled in this paper records USD 9.45 billion of depositor-facing gross losses across 1,075 operational-risk events on DeFi protocols between 2020-02-11 and 2026-05-29.

The decisive difference is *who bears that residual*. With no deposit insurance and no resolution authority, in DeFi the depositor, not the protocol, is on the hook. Faced with this heavy-tailed loss process, the two participants can each respond, and this paper is organized around their two responses.

*The protocol can hold a capital buffer.* Some protocols maintain governance-owned on-chain reserves with the operational-risk function Basel calls capital: Aave’s Umbrella safety module, the Sky surplus buffer, and Compound’s per-market reserve factor. No regulatory framework requires this; the buffers that exist are sized by governance vote rather than by any loss model.

*Where the protocol holds no buffer, the depositor should demand a risk premium.* By the design of a lending market, depositors who bear more operational risk should require a higher supply yield through the borrow rate the market clears. Where that yield embeds a premium over the risk-free rate sized to the operational-risk tail, depositors bear the risk knowingly and are compensated for it: the same market-discipline mechanism by which uninsured bank creditors price institutional risk [15, 18]. Market discipline presumes the depositor can price the risk, which requires disclosure the retail depositor typically lacks, a gap the policy discussion returns to.

The two responses are substitutes for the same risk, made by different participants, and neither is disciplined by a regulator. This paper asks *whether either response is sized adequately for the operational-risk tail it is meant to cover*, and makes four contributions to answer it.

*Contributions.* First, I assemble the most comprehensive publicly-reproducible DeFi operational-risk event dataset I am aware of:  $n = 1,075$  deduplicated events and USD 9.45 B of gross losses over 2020–2026, from seven public sources, filtered to DeFi protocols and tagged by sub-sector and Basel Level-1 event type [4]. Second, I quantify the loss process as a bank supervisor would, fitting a per-sector LDA and benchmarking the tail against banking operational risk [10, 19, 22]; the four core DeFi sectors prove no heavier-tailed than banking, while Bridge, Derivatives, and the residual Other sector exhibit cyber-loss-level tails ( $\xi \approx 1.6$ ; cf. [16]) with point estimates past the infinite-mean boundary. Third, I test the protocol’s response (capital buffers) and the depositor’s (risk premia) against that common benchmark. Fourth, I find the two margins move together in the right direction: venues holding a buffer pay a significantly lower premium than those without (Mann–Whitney  $p = 0.01$ ), evidence the market prices operational

risk, but at an order of magnitude below the modeled tail. I close with policy recommendations.

*Organization.* Section 2 situates the contribution within the literature, Section 3 documents the data and Basel taxonomy, and Section 4 sets out the LDA methodology. Section 5 reports the empirical per-sector severity, frequency, and capital fits, and Section 6 tests the protocol’s and the depositor’s responses and whether they substitute. Section 7 discusses the substitution between them, implications, policy, and limitations, and Section 8 concludes.

## 2 Related Work

This paper sits at the intersection of three empirical literatures: operational-risk modeling in banking, empirical DeFi security, and deposit pricing and market discipline.

*Operational risk in banking.* The statistical study of operational-risk losses begins with Moscadelli [22], who fit *generalized Pareto distributions* (GPD) to the Basel Committee’s 2002 loss-data exercise and reported tail indices  $\hat{\xi} \in [0.85, 1.39]$  across the eight business lines, establishing heavy tails with infinite higher moments. de Fontnouvelle et al. [19] found tail indices near or above one in internal-LDA models, with aggregate-loss *value-at-risk* (VaR) varying by an order of magnitude across severity choices. Cope et al. [10] catalogued the resulting pitfalls and Chernobai et al. [9] gave the canonical textbook treatment. Cyber risk is the closest non-financial analogue: GPD cyber-loss fits reach  $\hat{\xi} \approx 1.60$ , most subcategories in the infinite-mean regime [16], with the tail itself evolving over time [31]. The Basel Committee ultimately replaced the internal-model Advanced Measurement Approach (AMA) with the standardized approach, citing model complexity and excessive capital variability [5]. My DeFi findings reproduce that pathology.

*Empirical DeFi security.* The DeFi-security literature is younger but developing rapidly. Werner et al. [30] and Zhou et al. [32] provide systematization-of-knowledge surveys of the attack surface; the Zhou corpus of 181 incidents to April 2022 is closest in spirit to my dataset, though qualitative rather than tail-fitting. Qin et al. [24] characterize flashloan-enabled attacks and price-oracle manipulation, the mechanisms behind much of the Lending and Stablecoin losses I measure; Perez and Livshits [23] show that most statically vulnerable contracts are never exploited, motivating the ex-post realized-loss measurement I adopt. On the classification side, Weingärtner et al. [29] propose a “risk wheel” that sorts DeFi risks into systematic and unsystematic classes, and Arora et al. [2] build an FMEA risk-scoring framework—both *ex ante* qualitative frameworks, complementary to the *ex post* quantitative fit here. What this literature does not provide, and what I add, is a deduplicated multi-source event dataset tagged to the banking operational-risk taxonomy and fitted per sector.

*Yield pricing and market discipline.* The yield leg draws on bank market discipline. Flannery [18] reviews evidence that uninsured-creditor yields embed bank-condition information, and Egan et al. [15] formalize the joint determination

of deposit yields and capital as substitutes. For DeFi, Heimbach et al. [20] show Uniswap V3 LP fees often fail to compensate for impermanent loss, and Cornelli et al. [11] document that DeFi lending supply is driven by search-for-yield rather than risk-based pricing. Both legs have been studied separately; the substitution between protocol capital and depositor yield, which I estimate directly, has not.

*The gap.* Three gaps emerge. The DeFi-security literature classifies risks and surveys incidents but publishes no deduplicated, multi-source, event-level loss dataset tagged to the banking operational-risk taxonomy; neither it nor the cyber-loss literature reports per-sector severity and frequency parameters for DeFi; and no work places bank market discipline and operational-risk capital side by side as competing responses to a single quantified DeFi loss tail. This paper closes all three.

### 3 Data: A Consolidated DeFi Operational-Risk Dataset

A credible LDA fit requires an extensive event-level loss dataset. Previous DeFi-security and crypto-crime studies rely on a single source (DefiLlama, SlowMist, Chainalysis, or a curated list), each with known coverage gaps. I merge seven public feeds, deduplicate across them, and tag every record against a DeFi sector and the Basel Level-1 operational-risk taxonomy.

*Sources.* The seven feeds are (A) the DefiLlama hack ledger [13]; (B) the rekt.news leaderboard [25]; (C) the SunWeb3Sec *DeFiHackLabs* catalog [27]; (D) the kismpl23 *DeFi-Security-Incident* corpus [21]; (E) the BlockSec Security Incidents Library [6]; (F) the de.fi/rekt-database [12] (4,030 records, 2011–2026); and (G) the SlowMist Hacked tracker [26] (2,100 records, 2012–2026). Sources (F) and (G) are broadest and include out-of-scope memecoin honeypots, CEX failures, and individual-user phishing, removed by the filter below.

*Consolidation.* Feeds are normalized to a common schema and deduplicated in two passes: name clustering within a 21-day window, then a same-date ( $\pm 7$  d) and loss-within- $\pm 10\%$  pass that catches cross-named duplicates (“Ronin Bridge” / “Ronin Network”). The reconciled loss is the *median* of per-source amounts. A regex removes out-of-scope records (CEX events, memecoins, NFTs, custodial-wallet failures, Ponzi schemes, unattributed phishing, wallet-software exploits), and a second curated pass drops records that the regex missed through an explicit exclusion list. A total of 153 events is excluded as Non-DeFi this way.

*Sector inference.* Each event is assigned to one of seven sub-sectors: *Bridge* (cross-chain message-passing and token-bridging), *Lending* (permissionless money markets), *DEX* (automated market makers and order-book exchanges), *Yield* (aggregators, vaults, and yield-tokenization), *Stablecoin* (algorithmic, collateralized, and basis-backed), *Derivatives* (perpetual and option DEXs, synthetic-asset issuers, and liquid-staking / restaking tokens), and *Other* (residual DeFi protocols). The grouping follows the DeFi financial-primitives typologies of Werner et al. [30] and Auer et al. [3]. DefiLlama’s category field is used where present, else a regex chain resolves Bridge first, then Lending, DEX, Stablecoin, Yield,

**Table 1.** Basel Level-1 operational risk event types and their interpretation in a DeFi protocol

| Code | Basel Level-1                         | DeFi operational reading                                                                                                                                           |
|------|---------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IF   | Internal Fraud                        | Rugpull, owner-drain, insider exit-scam, upgradeable-contract abuse, honeypot-as-protocol.                                                                         |
| EF   | External Fraud                        | Smart-contract code bugs (reentrancy, access-control, rounding), credential compromise (key phishing), and infrastructure attacks (DNS hijack, approval phishing). |
| CPBP | Clients, Products, Business Practices | Economic-design and improper-practice attacks: flashloan-governance, oracle and price manipulation, donate-to-reserves, share-price inflation.                     |
| BDSF | Business Disruption, System Failures  | Chain-level events: maximal extractable value (MEV), sequencer outage, consensus issue, chain halt.                                                                |
| EDPM | Execution, Delivery, Process Mgmt     | Team-side configuration and deployment errors: oracle mis-deployments, broken governance parameters.                                                               |

and Derivatives. A curated per-sector audit corrects residual mislabels (including DefiLlama tags that a protocol’s identity contradicts).

*Risk-category inference.* Every record is tagged with a Basel Level-1 event type, the canonical banking taxonomy defined in Basel II [4] and carried into the Basel III finalization. The five types that transfer to an autonomous protocol are read as in Table 1. Two additional categories, Employment Practices and Workplace Safety (EPWS) and Damage to Physical Assets (DPA), have no measurable on-chain analogue and are omitted. Textual evidence (technique, description, name) is applied first, with DefiLlama’s classification as fallback, so an operational signature overrides a generic “contract bug” label.<sup>1</sup>

*Coverage and window.* The analysis window runs from 2020-01-01 [1] to the 2026-05-29 data cutoff, giving the 1,164-event consolidated DeFi dataset of Table 2 (the earliest in-sample event is dated 2020-02-11). A depositor-facing filter then removes 89 events (USD 0.84 B) whose losses are not borne by the sector’s earn user (e.g., governance-token mints, over-liquidation, treasury drains), yielding the 1,075-event working sample (USD 9.45 B), of which 536 (50%) carry cross-source confirmation.

## 4 Methodology: The Loss-Distribution Approach

I quantify the DeFi operational-loss process per Sector with the standard operational risk loss-distribution approach (LDA), the same tooling a bank supervisor

<sup>1</sup> In the February 2026 Moonwell cbETH event an oracle priced collateral at USD 1.12 instead of  $\approx$  USD 2,200, and liquidation bots seized USD 1.78 m in bad debt within minutes. The source issue was an oracle misconfiguration (a team-side deployment error, EDPM), not a contract exploit (EF). See the Moonwell incident summary, <https://forum.moonwell.fi/t/mip-x43-cbeth-oracle-incident-summary/2068>.

**Table 2.** Per-source contribution: raw record count and the number entering the consolidated 1,164-event DeFi dataset (after removing non-DeFi records; deduplication yields fewer entries than the feeds’ sum)

| Source                      | Raw records In dataset |     |
|-----------------------------|------------------------|-----|
| DefiLlama hack ledger       | 528                    | 379 |
| kismp123 DSI corpus         | 823                    | 290 |
| BlockSec Incidents Library  | 285                    | 201 |
| rekt.news leaderboard       | 281                    | 219 |
| DeFiHackLabs catalog        | 615                    | 163 |
| de.fi/rekt-database         | 4,030                  | 704 |
| SlowMist Hacked             | 2,100                  | 639 |
| Consolidated (deduplicated) | 1,164                  |     |

applies under Basel. The LDA has three parts: a severity model for individual event losses, a frequency model for how many events occur per year, and their compound aggregation into an annual loss distribution. I follow Basel’s original AMA and price the tail with a 99.9% VaR of that distribution [5], adopted for comparability with the banking benchmark.

*Severity.* I treat individual event gross losses as independent and identically distributed (iid) draws from a severity distribution  $F_X$  and model its upper tail by *peaks-over-threshold* (POT). For a threshold  $u$  and excesses  $Y_i = X_i - u \mid X_i > u$ , the GPD has density

$$g_{\xi, \beta}(y) = \frac{1}{\beta} \left(1 + \xi \frac{y}{\beta}\right)^{-(1/\xi+1)}, \quad y > 0, \beta > 0, \xi \in \mathbb{R}, \quad (1)$$

fitted by maximum likelihood with multi-start Nelder–Mead. The threshold quantile  $q^*$  is chosen per sector by a plateau-stability rule on the grid  $\mathcal{Q} = \{0.50, 0.55, \dots, 0.90\}$  (the highest  $q$  at which  $\hat{\xi}(q)$  is stable to within  $\tau = 0.30$ ), subject to two practitioner floors on the exceedance count  $n_u$ :  $n_u \geq 20$  (Embrechts et al. [17]) and  $n_u/n \geq 10\%$  (the DuMouchel [14] “top 10%” rule) and falling back to the lowest eligible quantile otherwise.

*Frequency.* For each sector I fit a *negative binomial* (NB) count model  $N_s \sim \text{NB}(\mu_s, \alpha_s)$  to the monthly event series. I chose the NB over a Poisson model because the monthly event counts are strongly over-dispersed, and a per-sector likelihood-ratio test rejects Poisson in five of the seven sectors (Section 5). The NB is fitted at monthly frequency. Annual counts entering the compound are then derived as the sum of twelve i.i.d. monthly draws with dispersion  $\hat{\alpha}_{\text{annual}} = \hat{\alpha}_{\text{monthly}}/12$ . Severity and frequency are treated as independent, following standard banking-LDA practice [9, 10]. A detailed analysis of this assumption and a potential joint model of clustered severities is left to future work.

*Compound aggregation.* The annual aggregate loss is

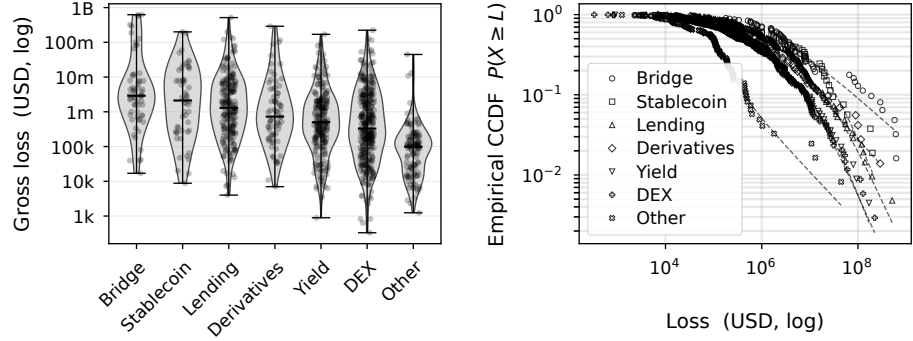
$$S_{\text{year}} = \sum_{i=1}^N X_i, \quad N \sim \text{NB}(\hat{\lambda}_{\text{annual}}, \hat{\alpha}), \quad X_i \stackrel{\text{iid}}{\sim} F_X, \quad (2)$$

where  $\hat{\lambda}$  is the per-sector annual event rate,  $\hat{\alpha}$  the NB dispersion, and  $F_X$  the piecewise empirical-body / GPD-tail mixture. Each simulated loss is capped at the largest single-protocol exposure in the sector, since one operational event cannot lose more than the assets at risk in one protocol. I draw  $2 \times 10^5$  simulated years per sector. The resulting mean  $\mathbb{E}[S]$  is the actuarial *pure premium*, and  $\text{VaR}_{99.9}$  is the *capital buffer* figure the Basel AMA would compute. I express each sector's  $\mathbb{E}[S]$  and  $\text{VaR}_{99.9}$  as a fraction of its trailing-365-day total value locked (TVL). TVL is the natural exposure base for these events because an admin-key compromise, contract exploit, or economic-design manipulation drains the smart contract's actual holdings, not just the collateral pledged against active positions. In other words, every dollar locked in the contract at the moment of the exploit is at risk. TVL is thus the on-chain analogue to a bank's assets on books rather than its risk-weighted assets. Two sectors lack a clean supply-side TVL, so I substitute the closest exposure base: circulating supply for Stablecoin, and residual TVL (total DeFi minus the five named sectors) for Other, both approximate and not strictly comparable to the rest.

*Per-protocol allocation.* No single protocol records enough events for its own fit, and pooling losses across protocols of very different size is a recognized challenge in operational-risk measurement [10]. Basel practice for banks lacking sufficient internal-loss data prescribes the same fallback: a business-line-level severity fitted collectively and allocated to the individual bank by an exposure indicator [9]. In the spirit of credibility theory [8], I therefore fit once per sector and treat its protocols as sharing that loss process per unit of TVL, a collective estimate toward which data-poor venues are shrunk. Let  $\text{TVL}_p$  be the TVL in protocol  $p$ ,  $\text{TVL}_s = \sum_{p \in s} \text{TVL}_p$  its sector total, and  $\lambda_s$  the sector event rate. The sector process then thins to each protocol. A sector event strikes protocol  $p$  with probability  $\text{TVL}_p / \text{TVL}_s$ , so  $p$  inherits a lower rate  $\lambda_p = (\text{TVL}_p / \text{TVL}_s) \lambda_s$  and the same per-event severity, capped at its own exposure. Were severity uncapped the mean would thin linearly and the pure premium would allocate by TVL share,  $\mathbb{E}[S]_p = (\text{TVL}_p / \text{TVL}_s) \mathbb{E}[S]_s$ ; the cap breaks that invariance, so  $\mathbb{E}[S]_p$  is simulated per venue and  $\mathbb{E}[S]_p / \text{TVL}_p$  falls as a venue shrinks. The tail departs from linearity more sharply. With  $S_p$  the annual aggregate loss of protocol  $p$ , a bad year is essentially one large event, and the single-loss approximation  $\text{VaR}_{99.9}(S_p) \approx F_X^{-1}(1 - 0.001/\lambda_p)$  [7, 17] scales as  $\text{TVL}_p^{\hat{\xi}}$  with  $\hat{\xi} < 1$ , so the VaR-to-TVL ratio rises as a venue shrinks. I therefore simulate  $\text{VaR}_{99.9}$  per protocol rather than scale the sector ratio, which understates the smaller venues.

## 5 Empirical LDA Fitting

*Data summary.* Losses are heavily right-skewed and differ sharply across sectors, motivating a per-sector LDA. The aggregate mean is USD 8.79 m against a



(a) Operational loss distribution; violins encode density, bars the median. (b) Operational loss tails (Log-log CCDF), with fitted POT-GPD (dashed)

**Fig. 1.** DeFi operational loss distribution and severity-tail fit across sectors

**Table 3.** Aggregate gross loss (USD m) by sector  $\times$  Basel Level-1 type. The largest cell is Bridge $\times$ EF (33% of total). Entries and margins are rounded independently.

| Sector \ type | EF    | IF    | CPBP  | EDPM | BDSF | Total |
|---------------|-------|-------|-------|------|------|-------|
| Bridge        | 3,084 | 127   | 6     | 8    | 12   | 3,237 |
| Lending       | 704   | 568   | 928   | 14   | 8    | 2,222 |
| Stablecoin    | 178   | 105   | 456   | 27   | 8    | 773   |
| Yield         | 430   | 267   | 206   | 17   | 6    | 926   |
| DEX           | 1,044 | 136   | 221   | 0    | 1    | 1,402 |
| Derivatives   | 148   | 0     | 275   | 2    | 380  | 805   |
| Other         | 35    | 4     | 48    | 0    | 0    | 87    |
| Total         | 5,623 | 1,207 | 2,140 | 68   | 415  | 9,452 |

median of USD 0.50 m and a maximum of USD 624 m (Ronin). Sector totals span two orders of magnitude, from Bridge (USD 3.24 B,  $n = 61$ ) to Other (USD 0.09 B,  $n = 121$ ). The per-sector distributions in Figure 1a differ in location and shape, Bridge carrying the highest median event loss (USD 2.90 m) from the large pooled collateral cross-chain bridges hold. The dominant loss mechanism differs too: sector and Basel event type are orthogonal, and crossing them (Table 3) shows Bridge and DEX as near-pure EF stories (locked collateral value), Lending and Stablecoin as CPBP (economic-design) failure heavy, and Derivatives as the only sector with significant BDSF exposure.

*Severity fits.* In Figure 1b the fitted GPD follows the empirical CCDF across the exceedance range in every sector, excesses scattering on both sides of the fit; in the  $\hat{\xi} > 1$  sectors it turns up faster than the data at the extreme top, the divergence the exposure cap addresses below. DeFi operational loss severity is thus heavy-tailed and the banking literature’s POT-GPD model applies. Table 4 reports the per-sector POT-GPD fit: Bridge ( $\hat{\xi} = 1.87$ ), Derivatives ( $\hat{\xi} = 1.45$ ),



**Table 4.** Per-sector POT-GPD fits with 1,000-replicate bootstrap CIs on  $\hat{\xi}$  at the plateau-selected quantile  $q^*$  ( $n_u$  exceedances); the last two columns give the Vuong non-nested test of the GPD against a lognormal, both fitted by maximum likelihood to the same exceedances ( $V > 0$  favors the GPD;  $|V| > 1.96$  separates them at 5%).

| Sector      | $n$ | $q^*$ | $n_u$ | $\hat{\xi}$ [95% CI] | $\hat{\beta}$ (USD m) | $V$   | $p$  |
|-------------|-----|-------|-------|----------------------|-----------------------|-------|------|
| Bridge      | 61  | 0.65  | 20    | +1.87 [+0.58, +3.18] | 17.6                  | -1.37 | 0.17 |
| Lending     | 207 | 0.90  | 21    | +0.75 [-0.28, +1.50] | 24.8                  | -1.32 | 0.19 |
| Stablecoin  | 52  | 0.60  | 21    | +0.65 [-0.39, +1.33] | 13.2                  | -0.03 | 0.97 |
| Derivatives | 71  | 0.65  | 25    | +1.45 [+0.42, +2.38] | 3.7                   | -1.04 | 0.30 |
| Yield       | 222 | 0.90  | 23    | +0.61 [-0.22, +1.24] | 11.5                  | -0.74 | 0.46 |
| DEX         | 341 | 0.85  | 51    | +0.72 [+0.18, +1.15] | 7.1                   | -0.02 | 0.98 |
| Other       | 121 | 0.80  | 24    | +1.58 [+0.54, +2.63] | 0.2                   | +0.57 | 0.57 |

and Other ( $\hat{\xi} = 1.58$ ) fit above the infinite-mean boundary. The four core sectors (Lending, Stablecoin, Yield, DEX) cluster tightly in  $\hat{\xi} \in [0.61, 0.75]$ , no heavier than the Moscadelli (2004) banking band  $[0.85, 1.39]$  and all finite-mean. With small exceedance samples in every sector ( $20 \leq n_u \leq 51$ ), these point estimates have to be interpreted with caution, as the wide confidence intervals show. I thus ask whether the lognormal model—an alternative severity model of the banking operational-risk literature [9, 19]—would offer a better fit, and report the Vuong [28] non-nested test (evaluated at the POT threshold  $u^*$ ) in Table 4. The test statistic returns a tie in every sector:  $|V| \leq 1.37$  against a 5% critical value of 1.96, with  $p \geq 0.17$  throughout. The lognormal therefore does not fit these exceedances better than the GPD does, and at these exceedance counts the data do not separate the two families at all. I therefore retain the GPD model.

*Frequency fits.* The monthly counts are strongly over-dispersed (cross-sector dispersion index  $D = 3.27$ )—events cluster rather than arrive independently—so a Poisson model would understate the annual aggregate and its  $\text{VaR}_{99.9}$ . Instead, I use the negative binomial (NB) model; per-sector fits are reported in Table 5 and give annual event rates from 8.3/yr (Stablecoin) to 54.2/yr (DEX). The table further reports results of a per-sector likelihood-ratio (LR) test that rejects Poisson at  $p \leq 0.02$  in five of seven sectors; it does not reject for Stablecoin ( $p = 0.59$ ), the sector with the fewest events, and for Derivatives ( $p = 0.05$ ) rejects it almost at  $p \leq 0.05$ , confirming NB is overall the better choice.

*Per-sector capital.* Combining severity and frequency, the compound LDA gives the per-sector capital buffer  $\text{VaR}_{99.9}$  of Table 6. Each simulated single-event loss is capped at the largest single-protocol exposure in its sector, since one operational event cannot drain more than the assets at risk in one protocol. The core sectors Lending, DEX, Yield, and Stablecoin produce  $\text{VaR}_{99.9}$  ratios of 14–18% of their exposure bases, well above the operational-risk capital banks hold [19]. Bridge, Derivatives and Other carry the largest requirements (23–32% of TVL), but only because each event is capped at exposure. Their  $\hat{\xi} > 1$  point estimates put severity in the infinite-mean regime, where no stable quantile exists,

**Table 5.** Per-sector negative-binomial frequency fits.  $n$  is the sector’s event count;  $\hat{\mu}_s$  the fitted mean monthly count and  $\hat{\lambda}_s = 12\hat{\mu}_s$  the implied annual event rate entering the compound;  $\hat{\alpha}_s$  the monthly NB dispersion. LR and  $p$  give the likelihood-ratio statistic against a Poisson and its  $p$ -value ( $\chi^2_1$ ).

| Sector      | $n$ | $\hat{\mu}_s$ | $\hat{\lambda}_s$ | $\hat{\alpha}_s$ | LR   | $p$                   |
|-------------|-----|---------------|-------------------|------------------|------|-----------------------|
| Bridge      | 61  | 0.80          | 9.7               | 0.72             | 6.2  | 0.013                 |
| Lending     | 207 | 2.72          | 32.9              | 0.26             | 10.2 | 0.0014                |
| Stablecoin  | 52  | 0.68          | 8.3               | 0.14             | 0.3  | 0.590                 |
| Derivatives | 71  | 0.93          | 11.3              | 0.41             | 3.7  | 0.054                 |
| Yield       | 222 | 2.92          | 35.3              | 0.60             | 42.3 | $7.7 \times 10^{-11}$ |
| DEX         | 341 | 4.49          | 54.2              | 0.31             | 38.5 | $5.4 \times 10^{-10}$ |
| Other       | 121 | 1.59          | 19.2              | 0.89             | 30.4 | $3.6 \times 10^{-8}$  |

**Table 6.** Per-sector LDA capital ( $\text{VaR}_{99.9}$ , % of trailing-365d TVL), compounded from the severity and frequency fits.  $\text{VaR}_{99.9}^U$  is the same simulation with the exposure cap removed, and IQR the parametric-bootstrap interquartile range (200 replicates from the fitted body+GPD mixture, refit and simulated at the same  $q^*$ ).  $^\dagger$ circulating supply,  $^\ddagger$ residual TVL.

| Sector      | $n$ | TVL (B)          | $\text{VaR}_{99.9}$ (%) | $\text{VaR}_{99.9}^U$ (%) | IQR (%)      |
|-------------|-----|------------------|-------------------------|---------------------------|--------------|
| Bridge      | 61  | 49.1             | 32.5                    | 67,326                    | [25.5, 40.8] |
| Lending     | 207 | 66.8             | 18.0                    | 22.3                      | [3.2, 18.9]  |
| Stablecoin  | 52  | 26.5 $^\dagger$  | 16.1                    | 16.1                      | [3.1, 30.1]  |
| Derivatives | 71  | 65.9             | 22.8                    | 638.0                     | [21.8, 27.1] |
| Yield       | 222 | 11.4             | 15.3                    | 29.5                      | [5.5, 17.3]  |
| DEX         | 341 | 14.5             | 14.1                    | 46.2                      | [10.5, 15.9] |
| Other       | 121 | 18.2 $^\ddagger$ | 27.5                    | 285.9                     | [4.8, 27.7]  |

so uncapped the Bridge  $\text{VaR}_{99.9}$  diverges to  $673 \times \text{TVL}$  (Table 6,  $\text{VaR}_{99.9}^U$ ). The heaviest-tailed, most consequential sectors are thus exactly the ones the model cannot pin down unaided, the instability that retired the Basel internal-model AMA in 2017 [5].

*Robustness.* Tail-index estimates are known to be sensitive to a handful of extreme observations [17]. This is confirmed in my sample: dropping the single largest event moves  $\hat{\xi}$  by 0.16–0.5 across sectors, most sharply in Lending ( $\hat{\xi} = 0.75 \rightarrow 0.35$ ). Under any drop-top rule up to 0.5%, however, every sector retains its below/above infinite-mean characteristic. A cap-sensitivity check (Table 6,  $\text{VaR}_{99.9}^U$  column) shows that on the net basis the exposure cap binds for the reported sectors. Removing it raises Lending from 18% to 22%, DEX from 14% to 46%, and Yield from 15% to 30% of TVL. Their  $\text{VaR}_{99.9}$  should therefore be read as cap-bounded, the 99.9% year being dominated by the largest venue’s at-risk funds failing. Only Stablecoin is cap-insensitive, its 99.9% loss falling below the single-protocol cap. Bridge and Derivatives ( $\hat{\xi} > 1$ ) are stable only under the cap. Without it the Bridge figure diverges to  $673 \times \text{TVL}$  and

**Table 7.** The four top-10 Lending venues holding a buffer (top-10 selected by gross supplied TVL, June 2026; TVL column is net, the funds a contract actually holds). Parentheses give buffer coverage of per-protocol  $\mathbb{E}[S]$  and  $\text{VaR}_{99.9}$ . Buffer data sources: <sup>a</sup>aave.com; <sup>b</sup>skyeco.com; <sup>c</sup>compound.woof.software; <sup>d</sup>bscscan.com.

| Protocol        | TVL (B) | Buffer (B)        | $\mathbb{E}[S]$ (B) | $\text{VaR}_{99.9}$ (B) |
|-----------------|---------|-------------------|---------------------|-------------------------|
| Aave V3         | 11.85   | 0.36 <sup>a</sup> | 0.08 (475%)         | 4.00 (9%)               |
| SparkLend       | 3.41    | 0.08 <sup>b</sup> | 0.02 (390%)         | 1.58 (5%)               |
| Compound V3     | 1.04    | 0.04 <sup>c</sup> | 0.01 (700%)         | 0.64 (6%)               |
| Venus Core Pool | 0.98    | 0.01 <sup>d</sup> | 0.01 (187%)         | 0.61 (2%)               |

Derivatives rises to 638%. The conclusions are also robust to the confidence level. Lowering the benchmark to 99.5% and 99% cuts the Lending sector capital from 18% to 7% and 5% of TVL and lifts buffered-venue mean coverage from 5% to 18% and 31%, so even at 99% the buffer covers only about a third of a requirement still an order of magnitude above bank operational-risk capital [19].

## 6 Bearing the Tail: Capital Buffers and Depositor Premia

An earn product’s operational-risk tail must be borne by someone. The protocol can fund it ex-ante by holding a capital buffer, or the depositor can be paid to bear it through a risk premium in the supply yield. I measure each margin against the per-sector  $\text{VaR}_{99.9}$  capital buffer benchmark of Section 5, then ask whether the two substitute.

*The protocol’s response.* The first response is to hold capital against the tail. Applying the per-protocol TVL-share allocation of Section 4 to the ten largest Lending venues by gross supplied TVL (DefiLlama [13], June 2026), four of the ten disclose an operational-risk reserve and six do not (Table 7). Under the per-sector homogeneity assumption, the four buffered venues cover on average only  $\sim 5\%$  of their modeled capital buffer (a 95% average shortfall), ranging from 2% (Venus) to 9% (Aave V3). The inadequacy is specific to the tail only though, all four buffers cover the pure premium (expected annual loss) at least once over, from Venus (187%) to Compound V3 (700%) (Table 7).

*The depositor’s response.* Where the protocol holds no buffer, or an inadequate one, the residual tail falls on the depositor, who should then be compensated ex-ante with a *risk premium*: a supply yield above the risk-free rate, the market-discipline mechanism by which uninsured creditors price institutional risk [15, 18]. I measure it as the trailing-30-day TVL-weighted stablecoin supply annual percentage yield (APY), minus the 3-month US Treasury bill rate of 3.70%<sup>2</sup>. This spread is only a proxy for the operational-risk premium, not an identified

<sup>2</sup> FRED series DTB3, for cross-jurisdiction comparability with the banking operational-risk literature [19, 22]. Pools that report a 0% supply APY are excluded. Supply APY includes both organic lending yield and incentives.

**Table 8.** Risk premium (30-day supply APY minus the 3.70% T-bill) for the ten largest Lending venues by buffer status. The  $\mathbb{E}[S]$  and  $\text{VaR}_{99.9}$  columns show pure premium and tail-risk premium; brackets give the risk premium’s coverage of each.

| Protocol           | Total yield | Risk premium | $\mathbb{E}[S]$ | $\text{VaR}_{99.9}$ |
|--------------------|-------------|--------------|-----------------|---------------------|
| <i>No buffer</i>   |             |              |                 |                     |
| Morpho Blue        | 4.60 %      | +90 bps      | 62 bps (144%)   | 3,905 bps (+2%)     |
| JustLend V1        | 3.42 %      | −28 bps      | 60 bps (−47%)   | 4,785 bps (−1%)     |
| Kamino Lend        | 2.91 %      | −79 bps      | 55 bps (−143%)  | 6,169 bps (−1%)     |
| Jupiter Lend       | 4.64 %      | +94 bps      | 55 bps (172%)   | 6,605 bps (+1%)     |
| Fluid Lending      | 6.27 %      | +257 bps     | 52 bps (497%)   | 6,507 bps (+4%)     |
| Euler V2           | 3.59 %      | −11 bps      | 47 bps (−23%)   | 8,131 bps (0%)      |
| <i>With buffer</i> |             |              |                 |                     |
| Aave V3            | 3.36 %      | −34 bps      | 64 bps (−53%)   | 3,378 bps (−1%)     |
| SparkLend          | 2.84 %      | −86 bps      | 60 bps (−143%)  | 4,634 bps (−2%)     |
| Compound V3        | 2.85 %      | −85 bps      | 55 bps (−155%)  | 6,101 bps (−1%)     |
| Venus Core Pool    | 2.09 %      | −161 bps     | 55 bps (−295%)  | 6,226 bps (−3%)     |

price, since the supply yield also reflects utilization, rate governance, protocol-idiosyncratic risk factors, and other non-operational factors.

Table 8 shows the premium is there in direction but not in size. Across the unbuffered venues it averages +54 basis points (bps), which on average just covers their own pure premium (55 bps mean  $\mathbb{E}[S]$ , mean coverage  $\approx 100\%$ ) but is a negligible fraction of the tail-risk premium the per-protocol  $\text{VaR}_{99.9}$  implies (34–81% of TVL). That average rests on six venues and hides wide dispersion: the premium runs from +257 bps (Fluid) to −79 bps (Kamino), coverage of the pure premium from +497% to −143% with a median of 60%, and three venues yield below the risk-free rate—a depositor can earn less than the T-bill while bearing an uncapped operational-risk tail.

*Both margins together.* Buffer coverage and risk premium are the two margins at which the same tail can be funded. Because the two substitute, an efficient market would trade one for the other, buffered venues paying lower yield. The data run in that direction. Venues with a buffer pay a significantly lower premium than those without, a median −86 bps against +39 bps. The direction is right and statistically significant (one-sided Mann–Whitney  $p = 0.01$ ), but the scale is not. The whole premium cross-section spans just 418 bps, while the tail demands capital of about 18% of TVL, so both margins together leave it overwhelmingly unfunded, neither within an order of magnitude of pricing it. The directional gap survives extending the sample to all Lending venues with net TVL  $\geq$  USD 100 m ( $n = 19$ , one-sided  $p = 0.03$ ), and attenuates below, where posted yields are dominated by airdrop and other incentive programs rather than by market-cleared compensation for risk-bearing: the long-tail depositor is chasing incentives, not pricing risk.

## 7 Discussion

*An earn product without the capital behind it.* DeFi earn products are backed by neither a buffer nor a risk premium sized to the tail. Although the four DeFi core sector severity tails are no heavier than banking’s, the implied capital is larger: e.g. for Lending the per-sector  $\text{VaR}_{99.9}$  is  $\sim 18\%$  of TVL, far above the operational-risk capital banks hold [19], because the 99.9% year is dominated by the largest venue’s at-risk funds failing. Bridge and Derivatives are larger still (32% and 23%) but cap-determined, as their  $\xi > 1$  severity is unquantifiable without the cap. Of the ten largest Lending venues, four hold a buffer, covering on average  $\sim 5\%$  of their modeled capital buffer. The six without a buffer carry a supply-yield premium averaging +54 bps, which on average matches their 55-bps pure premium, though across only six venues coverage of it ranges from  $-143\%$  to  $+497\%$ . I find evidence that the two responses substitute: buffered venues pay a significantly lower premium—the trade-off an efficient market would produce—but both margins are an order of magnitude too small.

*Who bears the shortfall.* The premium I measure is the posted supply APY, the rate a retail depositor sees and takes. Depositors are not homogeneous: Cornelli et al. [11] find that search-for-yield drives supply, retail deposits in particular, and that behavior differs by account size. A professional participant can assess the tail, negotiate bilateral terms,<sup>3</sup> and size or exit the position. A retail depositor sees only the posted rate and, lacking standardized disclosure, bears the unpriced tail disproportionately.

*Policy.* DeFi protocols are not regulated as banks are, and mandating bank-style operational-risk capital is harder to implement in a world of decentralized, autonomous DeFi protocols. What market discipline does require is information: depositors bearing the operational-risk tail should receive adequate disclosure to judge whether the yield compensates them. The LDA here supplies the content of such a disclosure, and I recommend three measures. First, *standardized loss-and-yield disclosure*, a Pillar 3 analogue that consolidates the private loss feeds (DefiLlama / SlowMist / de.fi) into a comparable per-sector loss history reported alongside the supply yield and its implied risk premium versus the pure premium. Second, *buffer-and-tail disclosure*: each venue’s buffer, the share of its  $\text{VaR}_{99.9}$  allocation covered, and the residual tail the depositor bears. Third, *periodic recalibration*, because the tail is non-stationary [5, 15, 18, 31]. The disclosure duty need not fall on the protocol alone: a decentralized protocol may have no responsible entity, so the obligation can attach to any service provider that gives users access (front-ends, wallets, aggregators, custodial “earn” products), the identifiable points at which a retail depositor actually enters the position.

*A hybrid calibration standard.* Basel’s operational-risk framework runs a two-tier model: standardized business-line coefficients for banks lacking internal data, and an internal-model option for those with enough loss history [5]. The same

<sup>3</sup> Professional market makers strike bespoke arrangements with protocols through governance, as in Wintermute’s market-making agreement with MakerDAO (<https://thedefiant.io/news/defi/wintermute-makerdao>).

hybrid fits DeFi: the per-sector LDA of Section 4 is the default, and any venue with dense enough own-loss history can substitute a protocol-specific model, disclosed and periodically recertified.

*Where the capital sits.* Existing on-chain buffers (Aave’s Umbrella, Sky surplus buffer, Compound’s reserve factor) all sit inside the same smart-contract system whose failure they insure, inheriting the same code-risk. Two decoupling patterns are possible: *protocol-maintained recovery pools* for the idiosyncratic component, and *cross-protocol recovery funds*—the on-chain analogue to post-2008 bank resolution funds—for the systematic component. Neither is standard practice today.

*Limitations.* Six limitations bound the results. *Incomplete data.* The analysis rests on publicly reported losses; sub-threshold and privately resolved incidents are missing, biasing the body of the severity distribution downward but leaving the large tail events that drive the capital figures intact. *Homogeneity.* With no protocol generating enough events for its own fit, I estimate one severity per sector and allocate it by TVL share, assuming a common loss process per unit of TVL. A larger or more complex venue may bear more risk per dollar, which a size-covariate or hierarchical model would refine. *Recovery.* The figures are gross of funds later returned; netting recoveries lowers the aggregate, so the capital estimates are conservative. *Confounding.* The lower premia at buffered venues are a descriptive association, not an identified buffer effect. Buffered venues are also the largest and oldest, so brand and perceived safety may drive the yield gap as much as the buffer. Perceived safety is itself an imperfect proxy: industry-standard security audits target code-level vulnerabilities and offer no protection against the IF and EDPM categories that account for 26% of Lending losses. *Premium identification.* The supply-yield spread is a reduced-form proxy, not an identified risk price; it also reflects utilization, protocol-idiosyncratic risk factors, and rate governance. A controlled panel regression is the natural extension. *Stationarity.* The 2020–2026 window is fit as stationary, but frequency has fallen from the early rugpull wave and severity has moved toward larger, more concentrated targets; a regime-split fit would sharpen the point-in-time estimate.

## 8 Conclusion

DeFi protocols expose depositors to operational risk (USD 9.45 B across 1,075 events since 2020) but, unlike banks, need to hold no capital against it, and absent a voluntary buffer the residual is left with the depositor. Calibrating a Basel LDA gives a common benchmark, and against it neither margin is sized to the tail: at the ten largest Lending venues the four disclosed buffers cover on average 5% of the modeled  $\text{VaR}_{99.9}$ , and unbuffered venues carry a supply-yield premium averaging +54 bps, short of the tail by more than an order of magnitude. The two move together in the right direction—the market prices the risk, but at a magnitude that leaves the tail overwhelmingly unfunded. Because these protocols are not bank-regulated, the remedy I draw is not capital mandates but

disclosure: standardized loss, yield, and buffer-coverage information, required of the protocol or any service provider that fronts access to it.

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